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Li_2MTeO_6 (M = Ti, Sn): Mid-Infrared Nonlinear Optical Crystal with Strong Second Harmonic Generation Response and Wide Transparency Range

Xiaoli Du[+], Xiaojie Guo[+], Zeliang Gao, Fuan Liu, Feifei Guo, Siyuan Wang, Haoyuan Wang, Youxuan Sun, and Xutang Tao**

Abstract: *Oxide crystals have been widely used in nonlinear optics (NLO) in the ultraviolet–visible and near-infrared regions. Most traditional oxide crystals are restricted to the mid-infrared region due to their narrow transmission window. Hence, attempting to extend infrared cutoff wavelength of oxides has attracted much attention. Herein, we report two new tellurates Li_2TiTeO_6 (LTT) and Li_2SnTeO_6 (LST) with broad transparent regions of 0.38–6.72 and 0.38–6.86 μm, respectively, as excellent candidates for mid-infrared NLO applications. Both LTT and LST crystallize in the orthorhombic space group Pnn2. The LTT crystal exhibits intense powder second-order generation efficiency (26 × KDP) under the fundamental wavelength of 1064 nm. First-principles calculations and dipole moments were used to illustrate the results of the powder second-harmonic generations based on the crystal structures. Our results provide a novel oxide NLO crystal with a strong SHG and wide transparency range. They also pave a way for the design of new oxide mid-IR NLO crystals.*

Introduction

Mid-infrared (MIR) nonlinear optical (NLO) crystals with the capability of completely covering the range of 3–5 μm by frequency conversions play a crucial role in military and civil fields such as military communication, laser lithography, medical treatment, resource detection, etc.[1] Over the past decades, MIR NLO crystals have been extensively studied and mainly focused on non-oxide systems such as chalcogenides and pnictides, including the successfully commercialized crystals $AgGaS_2$, $AgGaSe_2$, and $ZnGeP_2$.[2] However, these crystals are hindered regarding their extensive applications due to their low laser damage thresholds (LDTs), strong two-photo absorption, non-phase-matching (NPM) feature, and the difficulty of large-sized crystal growth attributed by stoichiometric variation.[3] Oxide crystals usually exhibit wider band gaps and larger LDTs than chalcogenides and pnictides, which are important for NLO crystals. However, it is difficult to completely cover the atmospheric window of 3–5 μm for most oxide crystals due to the absorption of high frequency phonons or their overtones from the light atoms. The IR cutoffs of some oxide crystals, such as $KTiOPO_4$ (KTP), LiB_3O_5 (LBO), β-BaB_2O_4 (BBO), and $LiNbO_3$ (LN), cannot reach 5 μm.[4] The traditional oxide crystals are limited to their narrow mid-IR transmission band. Therefore, the design of mid-IR NLO oxide crystals with wide mid-IR transmission band is challenging.

Under the guidance of the second-order Jahn–Teller (SOJT) effect, a series of compounds with d^0 transition metal cations (Mo^{6+}, W^{6+}, Ti^{4+}, Zr^{4+}) and cations with nonbonded electron pairs (Se^{4+} or Te^{4+}) have been synthesized and grown into single crystals such as β-$BaTeMo_2O_9$ (β-BTM), α-$BaTeMo_2O_9$ (α-BTM), $Na_2TeW_2O_9$, $Cs_2TeMo_3O_{12}$, $Cs_2TeW_3O_{12}$, $MgTeMoO_6$, $CdTeMoO_6$.[5] The crystal properties show that not only the crystal structures have been adjusted to be non-centrosymmetric, but also the IR cutoffs have been extended to beyond 5 μm because of the presence of the tellurium. The β-BTM crystal, as a typical representative NLO molybdate crystal, exhibits a wide transparency range of 0.5–5 μm and the effective nonlinear coefficient of 10.3 pm V^{-1} at 1064 nm. As one of the molybdates, β-BTM possesses both excellent second-order NLO and stimulated Raman scattering properties. In our studies, the stimulated Raman scattering property might impact the second-order nonlinear application. The studies of the β-BTM series crystals indicate that combining special kinds of oxide-based systems and heavy atoms (Te) is an effective approach to adjust the NCS structure and extend the IR transmission range, which is a new route to developing high-performance MIR NLO crystals.[6]

In order to reduce the stimulated Raman scattering property, oxides containing Te and Ti elements have attracted our attention. The tellurates present a low phonon frequency which indicates a wide IR transmission window. And the distortion magnitude presents the situation of Mo > V > W > Ti > Nb > Ta > Zr[7] in the SOJT effect, which means that the introduction of Ti^{4+} cations should assist the structural distortions. The different titanium–oxygen bonds of the TiO_6 octahedron illustrate the d^0 transition metal atom Ti out of center, which would enlarge its polarizability to obtain higher intensity of the second harmonic generation (SHG) signal, as exhibited in the crystals of the KTP family, $BaTiO_3$, $Bi_4Ti_3O_{12}$, $PbTiO_3$, $Li_2K_4TiOSi_4O_{12}$, etc.[7,8] Lin and co-workers reported that Sn^{4+} with a d^{10} electronic structure has

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a large polar displacement that can result in structural distortions, and the SnO_6 octahedron that exhibits a larger flexibility index and the high flexibility of the Sn–O bond greatly enhance the SHG effect under the guidance of the "flexible dipole model" theory.[9] Since then, a number of compounds containing Sn^{4+} cations, such as $La_3SnGa_5O_{14}$, $SnB_2O_3F_2$, $SrZnSnS_4$, and $Sn(IO_3)_2F_2$,[9,10] have shown potential applications in the mid-IR region as NLO materials. Furthermore, Ti^{4+} cations and Sn^{4+} cations with smaller ionic radius belong to d^0 and d^{10} type metals, respectively, which would make the crystal exhibit a small unit cell volume and an increased density of polar groups to contribute a stronger SHG effect. Inspired by these design strategies, two new mid-infrared nonlinear optical crystals, Li_2TiTeO_6 (LTT) and Li_2SnTeO_6 (LST), were designed and synthesized in this work.

The polycrystalline powders of LTT and LST were synthesized by a solid state reaction. First-principles calculations and dipole moments were used to illustrate the results of the powder second harmonic generation based on the crystal structures. Then, single crystals of LTT and LST were grown by the top-seeded solution growth (TSSG) method for the first time. The LTT and LST crystals exhibit broad transmission ranges of 0.38–6.72 and 0.38–6.86 μm, respectively, and the LTT crystal exhibits a high SHG efficiency of 26 × KDP, which indicates that LTT crystals have potential applications in nonlinear optics.

Results and Discussion

Polycrystalline samples of Li_2MTeO_6 (M = Ti, Sn) were synthesized from the stoichiometric mixture of Li_2CO_3, TiO_2 (or SnO_2), and TeO_2 through a conventional solid-state reaction. The pure phases of Li_2MTeO_6 (M = Ti, Sn) were confirmed by XRD analysis. Figure 1 shows the experimental powder XRD patterns of Li_2MTeO_6 (M = Ti, Sn) polycrystalline samples compared with the standard PDF cards. By XRD comparison, the prepared powders are pure phases, as they are in good agreement with the calculated patterns.

The thermal stability of Li_2MTeO_6 (M = Ti, Sn) crystals was evaluated by TGA/DSC measurement. As shown in Figure 2a, there are two obvious endothermic peaks for LTT around 822 °C and 868 °C in the DSC curve; meanwhile, mass losses are observed in the TGA curve. When the LTT samples are heated at 1000 °C for 10 h, the compound decomposes to TiO_2 and Li_2TeO_3 according to the XRD pattern of the residues as shown in Figure S1a, which indicates that LTT is an incongruently melting compound. Meanwhile, there is a similar phenomenon in the TGA/DSC of LST. During the heating process, LST shows pronounced weight loss at 850 °C in the TGA curve, and an endothermic peak in the DSC curve as shown in Figure 2b, which means that LST is stable up to 850 °C. After the measurements, the residues were confirmed as SnO_2 and Li_2TeO_3, as indicated by the PXRD analysis in Figure S1b. The results suggest that LST melts incongruently and decomposes before the homogeneous solution is formed. Therefore, the TSSG method is a preferable way to grow LTT and LST crystals with high quality and large size. Moreover, there is no weight change when measuring the weight of Li_2MTeO_6 (M = Ti, Sn) crystals after soaking in distilled water for several days, indicating that Li_2MTeO_6 (M = Ti, Sn) crystals have good chemical stability.

Because of the incongruent melting characteristic of Li_2MTeO_6 (M = Ti, Sn) crystals, a modified TSSG should be adopted to grow Li_2MTeO_6 (M = Ti, Sn) crystals. Meanwhile, there are some advantages for Li_2MTeO_6 (M = Ti, Sn) crystals grown by the flux method, including euhedral habit and a low dislocation density. In addition, the selection of the flux in the process of crystal growth is very vital to gain high-quality crystals. TeO_2, as a flux, has been extensively exploited to grow single crystals because it has a low melting point and an excellent capability for dissolving oxides. And to some extent, the introduction of lithium ions can increase the crystal size. Hence, we chose the mixture of TeO_2 and Li_2CO_3 as the flux

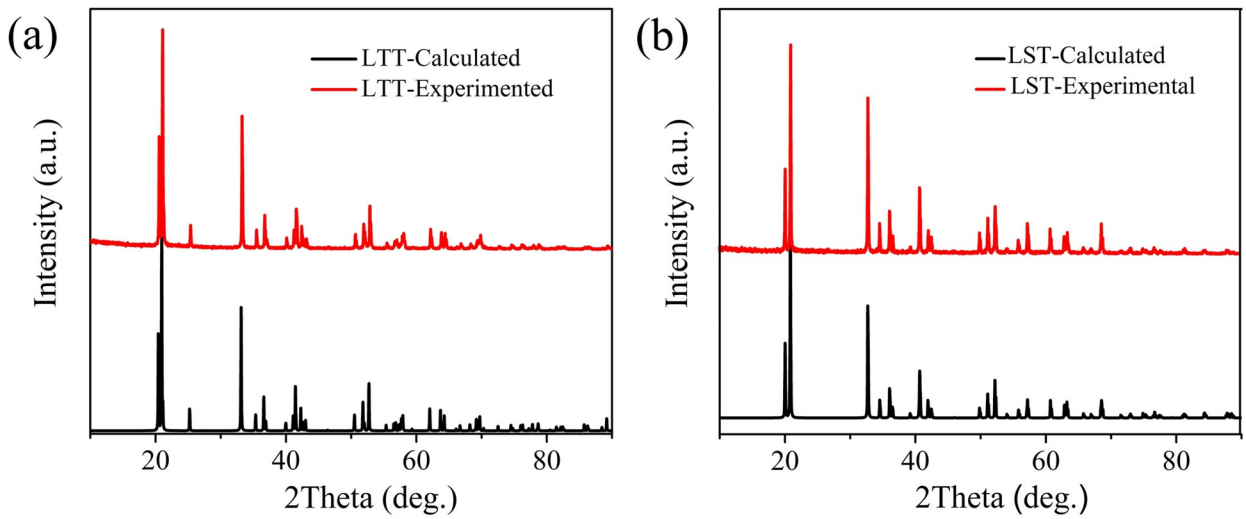

Figure 1. a) Experimental and calculated PXRD patterns of LTT polycrystalline samples. b) Experimental and calculated PXRD patterns of LST polycrystalline samples.

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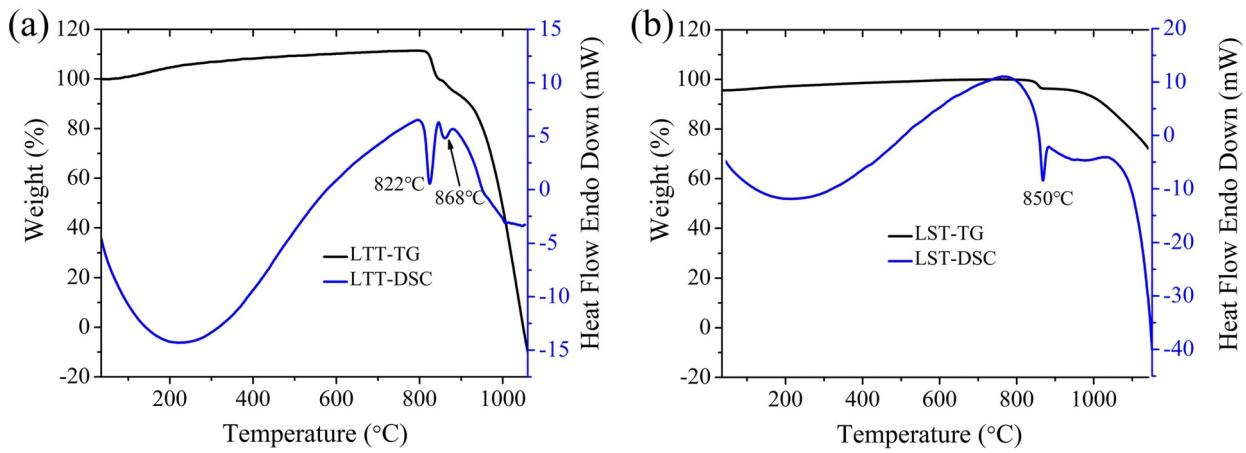

Figure 2. a) TGA and DSC curves of LTT polycrystalline samples. b) TGA and DSC curves of LST polycrystalline samples.

to obtain transparent and well-shaped faceted single crystals of Li_2MTeO_6 (M = Ti, Sn). Through previous explorations, the optimal flux ratios were explored for growing crystals. Ultimately, high-quality transparent Li_2MTeO_6 (M = Ti, Sn) crystals were obtained. More importantly, the crystal growth habit and environmentally friendly characteristic of Li_2MTeO_6 (M = Ti, Sn) are particularly advantageous for applications in the NLO field.

On the basis of single-crystal XRD analysis, LTT and LST are isostructural and crystallize in the orthorhombic space group *Pnn*2. The unit cell parameters of LTT and LST are listed as $a = 4.9015$ Å, $b = 5.0692$ Å, $c = 8.4083$ Å and $Z = 2$ for LTT and $a = 4.9292$ Å, $b = 5.2076$ Å, $c = 8.5290$ Å and $Z = 2$ for LST, respectively. The refined crystallographic data and details of the experimental condition for Li_2MTeO_6 (M = Ti, Sn) are shown in Table 1. The detailed crystallographic data of LTT and LST are listed in Tables S1–S6. As shown in Figure 3, the Li_2TiTeO_6 crystal structure features a three-dimensional framework composed of one type of Ti atoms, one kind of Te atoms, two different Li atoms and three differently coordinate O atoms. In its structure, the Ti atom is coordinated by six oxygen atoms to form a disordered TiO_6 octahedron, and Ti–O bond lengths are between 1.931 Å and 2.029 Å. Meanwhile, the Te^{6+} cations with d^{10} electronic configuration and polar displacement are also bonded to six oxygen atoms with Te–O bond lengths ranging from 1.892 Å to 1.948 Å and forming a disordered TeO_6 octahedron. In the structure of LTT crystals, the Ti–O and Te–O bond lengths are in the normal distance. Two TiO_6 octahedra and one TeO_6 octahedron are connected by edge-sharing along the *b*-axis to form a $[Ti_2TeO_{14}]$ unit. Meanwhile, two TeO_6 octahedra and a TiO_6 octahedron are connected by edge-sharing along the *b*-axis to form a $[Te_2TiO_{14}]$ unit. $[Ti_2TeO_{14}]$ units and $[Te_2TiO_{14}]$ units are linked with each other by corner-sharing along the *c*-axis to form six-membered rings. Two types of Li^+ cations are located in the six-membered rings' cavity to maintain charge balance. The cation ordering sequence is Ti-Te-Li-Ti-Te-Li… along the *c*-axis. With the same situation as shown in Figure 3, Sn–O bond lengths vary from 1.989 to 2.091 Å and Te–O bond lengths range from 1.876 to 1.980 Å in LST. SnO_6 octahedra and TeO_6 octahedra are connected by edge-sharing along the *a*-axis to form $[Sn_2TeO_{14}]$ units and $[SnTe_2O_{14}]$ units. The $[Sn_2TeO_{14}]$ units and $[SnTe_2O_{14}]$ units are linked to each other by corner-sharing along the *c*-axis with Li^+ cations residing in the tunnels to balance the valence state. The cation ordering sequence is Sn-Te-Li-Sn-Te-Li… along the *c*-axis. The octahedra share corners and edges to construct the three-dimensional framework holding the structure firmly together.

In order to clarify the asymmetrical coordination environments, the local dipole moments of the TeO_6 octahedron and TiO_6 octahedron in LTT and the TeO_6 octahedron and SnO_6 octahedron in LST were calculated by the bond-valence

Table 1: Crystal data and structure refinement parameters for Li_2MTeO_6 (M = Ti, Sn).

Empirical formula	Li_2TiTeO_6	Li_2SnTeO_6
Formula weight [g mol^{-1}]	285.38	356.17
T [K]	298(2)	300(2)
λ [Å]	0.71073	0.71073
Crystal system	Orthorhombic	Orthorhombic
Space group	*Pnn*2	*Pnn*2
a [Å]	4.9015(7)	4.9292(10)
b [Å]	5.0692(8)	5.2076(7)
c [Å]	8.4083(12)	8.5290(13)
V [Å^3]	208.92(5)	218.93(6)
Z	2	2
ρ_{calcd} [g cm^{-3}]	4.537	5.403
μ [mm^{-1}]	8.826	12.305
F(000)	256	312
Crystal size [mm^3]	0.078×0.081×0.082	0.049×0.057×0.059
Data/restraints/parameters	727/1/42	504/1/43
R(int)	0.0424	0.0569
GOF (F^2)	1.167	1.127
Final *R* indices [$I > 2\sigma(I)$][a]	$R1 = 0.0213$, $\omega R_2 = 0.0520$	$R1 = 0.0214$, $\omega R_2 = 0.0487$
R indices (all data)	$R1 = 0.0263$, $\omega R_2 = 0.0561$	$R1 = 0.0257$, $\omega R_2 = 0.0526$
Absolute structure parameter	0.02(8)	0.17(9)
CCDC	2103512	2103514

[a] $R_1 = \Sigma \| F_o| - |F_c\| / \Sigma |F_o|$ and $\omega R_2 = \{\Sigma[\omega(|F_o|^2 - |F_c|^2)^2] / \Sigma[\omega(|F_o|^4)]\}^{1/2}$.

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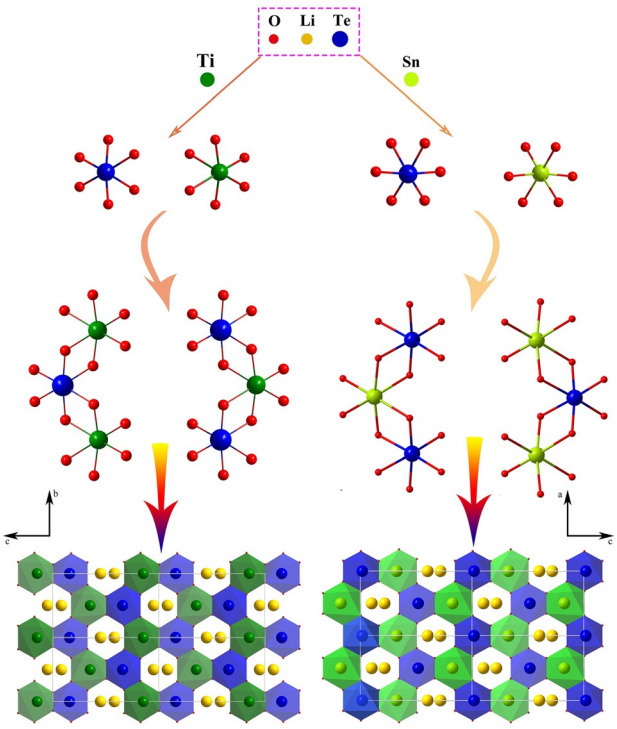

Figure 3. 3D framework structures for LTT and LST.

approach proposed by Poeppelmeier et al.[11] The magnitude of the distortion of the MO_6 octahedron can be determined by Equation (1).

$$\Delta_d = \frac{|(M-O_1)-(M-O_4)|}{|\cos\theta_1|} + \frac{|(M-O_2)-(M-O_5)|}{|\cos\theta_2|} + \frac{|(M-O_3)-(M-O_6)|}{|\cos\theta_3|} \quad (1)$$

$M-O_1$, $M-O_2$, $M-O_3$, $M-O_4$, $M-O_5$, and $M-O_6$ represent the individual bond lengths; $\cos\theta_1$ represents the angle between $M-O_1$ and $M-O_4$, $\cos\theta_2$ represents the angle between $M-O_2$ and $M-O_5$, and $\cos\theta_3$ represents the angle between $M-O_3$ and $M-O_6$. The direction and magnitude of the dipole moments of the TiO_6 octahedron and TeO_6 octahedron in LTT and SnO_6 octahedron and TeO_6 octahedron in LST are summarized in Table 2. With the dipole moments, the vector summation over the TiO_6 geometry shows a net dipole moment of 5.2594 Debye and the net dipole of the TeO_6 octahedron is 2.7841 Debye in LTT. The net dipole moment of the unit cell is 236.916×10^{-22} esu cm Å^{-3} in LTT. Meanwhile, the net dipole moment of the SnO_6 octahedron is 3.8361 Debye, the net dipole moment of the TeO_6 octahedron is 3.3292 Debye and the net dipole moment of the unit cell is 46.042×10^{-22} esu cm Å^{-3} in LST. The dipole moment vectors of the TeO_6 and TiO_6 octahedra are along the direction towards their local C_4[001] direction. The dipole moments of the TeO_6 and TiO_6 octahedra are parallel to the *c* crystallographic axis, however, their vectors are antiparallel as shown in Figure 4. There is a similar phenomenon for the TeO_6 and SnO_6 octahedra in LST. The phenomenon with opposite direction of polarized octahedron may cause a partial cancellation of the dipole moment. Nevertheless, LTT and LST crystals still exhibit macroscopic nonlinear optical effects.

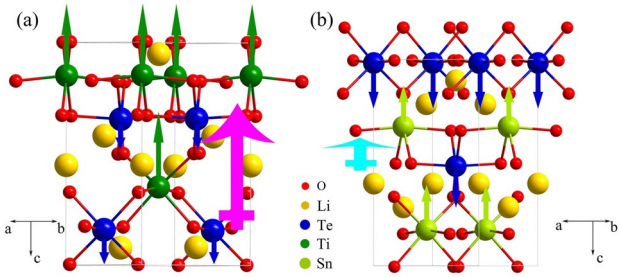

Figure 4. a) The dipole moment vectors of the LTT unit cell. b) The dipole moment vectors of the LST unit cell.

Both LTT and LST crystals belong to *Pnn*2, a non-centrosymmetric (NCS) space group, which is expected to exhibit excellent NLO properties. The curve of the powder SHG signals detected and recorded was plotted against the particle sizes in Figure 5. The SHG intensity of LTT and LST crystals increases along with the increase of the particle sizes before it attains the maximum value independently of the particle size. The result shows that LTT and LST crystals are type-I phase-matchable at the pump laser light of 1064 nm. Remarkably, the results show that the powder SHG intensity of the LTT crystal is about 26 times larger than that of KDP at 1064 nm in the same particle size ranging from 420 to 500 μm, as shown in Figure 5b. However, the SHG intensity of the LST crystal is 2.5 times that of KDP at 1064 nm in the same particle size range. According to the dipole moment analysis

Table 2: The direction and magnitude of the dipole moments of the TiO_6 and TeO_6 octahedra in LTT and the SnO_6 and TeO_6 octahedra in LST.

Crystal	Asymmetric units	Polyhedron distortion, Δ_d		Dipole moment				
		Direction	Magnitude	*x*(*a*)	*y*(*b*)	*z*(*c*)	Magnitude Debye	$\times 10^{-22}$ esu cm Å^{-3}
Li_2TiTeO_6	TiO_6	*C*4[001]	0.20	0.0000	0.0000	−5.2594	5.2594	251.74
	TeO_6	C_4[001]	0.06	0.0000	0.0000	2.7841	2.7841	133.24
	Unit cell	–	–	0.0000	0.0000	−4.9506	4.9506	236.916
Li_2SnTeO_6	SnO_6	C_4[001]	0.014	0.0000	0.0000	−3.8361	3.3861	206.07
	TeO_6	C_4[001]	0.094	0.0000	0.0000	3.3292	3.3292	152.07
	Unit cell	–	–	0.0000	0.0000	−1.008	1.008	46.042

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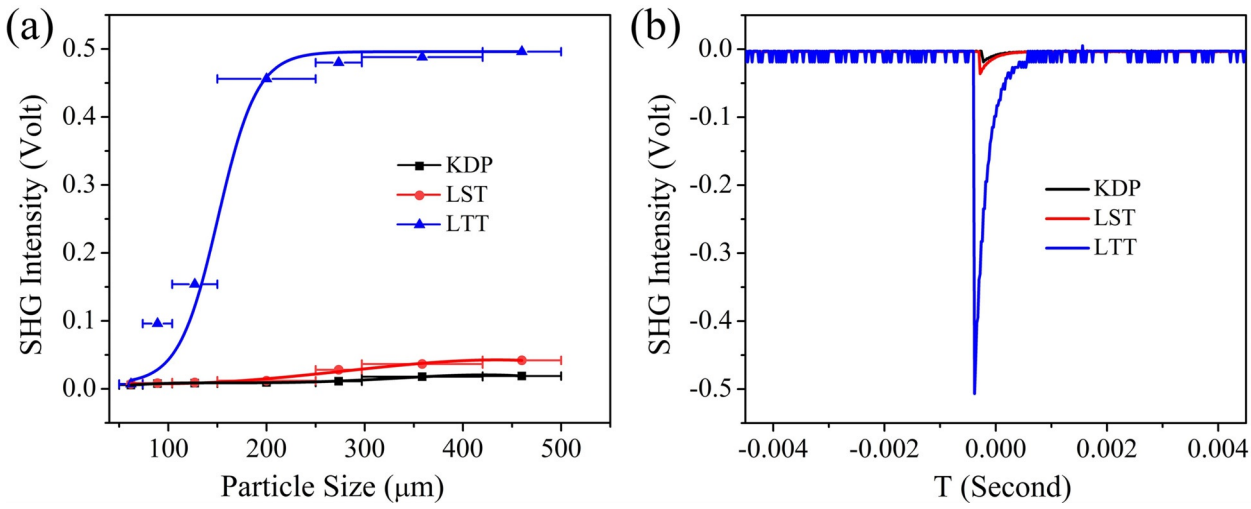

Figure 5. a) Phase-matching curve for Li_2MTeO_6 (M = Ti, Sn) (the red solid curve, the blue solid curve and the black solid curve are supposed to guide the eye and not a fit to the data). b) Oscilloscope traces of the SHG signals for Li_2MTeO_6 (M = Ti, Sn) and KDP in the same particle size of 420–500 μm.

of Li_2MTeO_6 (M = Ti, Sn), the magnitude of the net dipole moment of the LTT unit cell is five times that of LST, which is consistent with the measurement of these crystals. The length of the Ti–O bonds ranges from 1.931 to 2.029 Å, which plays an important role in the microscopic polarizability of LTT crystals. Based on the previous studies, the participation of Sn presents lower distortion of the MO_6 octahedron than Ti, which is consistent with our experimental results.[7,8b,12] It is worth noting that the SHG efficiency is related not only to the dipole moment of the unit cell but also to the rate of the unit cell dipole moment changing under the external field, and the latter effects would influence the NLO property, which still needs further exploration.

The optical transmission spectra of LTT and LST crystals were recorded in the wavelength range of 200–7500 nm to evaluate the optical properties. Figure 6 shows the UV and IR transmittance spectra of Li_2MTeO_6 (M = Ti, Sn) crystals at room temperature. The UV absorption edges of the Li_2TiTeO_6 and Li_2SnTeO_6 crystals are both located at 0.38 μm. With the increase of wavelengths, the transmittance of the Li_2MTeO_6 (M = Ti, Sn) increases rapidly. The IR absorption edge of LTT (101) facet is extended to 6.72 μm. For LTT (101) facet, it exhibits a transmission band of 0.38–6.72 μm with a 65% transmittance and an absorption peak at 3 μm, which is attributed to the inclusions of the crystal. With the optimization of crystal quality, the absorption peak at 3 μm would disappear and the transmission should be improved. There remains a relatively high transmittance (80%) from 600 nm to 6 μm and the IR absorption edge is extended to 6.86 μm for the LST (110) facet. Furthermore, the LST crystal exhibits a high mid-IR transparency up to 80% and almost no absorption in the atmospheric transmission window (3–5 μm). The effective transmittance band of LST crystals covers a range of up to 5.50 μm with 50% transparency, which is superior to many other molybdenum/tellurite oxide crystals, including α-$BaTeMo_2O_9$ (0.4–5.4 μm), $Cs_2TeMo_3O_{12}$ (0.43–5.38 μm), $MgTeMoO_6$ (0.36–5.2 μm), $Na_2Te_3Mo_3O_{16}$ (0.42–5.4 μm), $Cs_2TeW_3O_{12}$ (0.41–5.31 μm). Moreover, as shown in

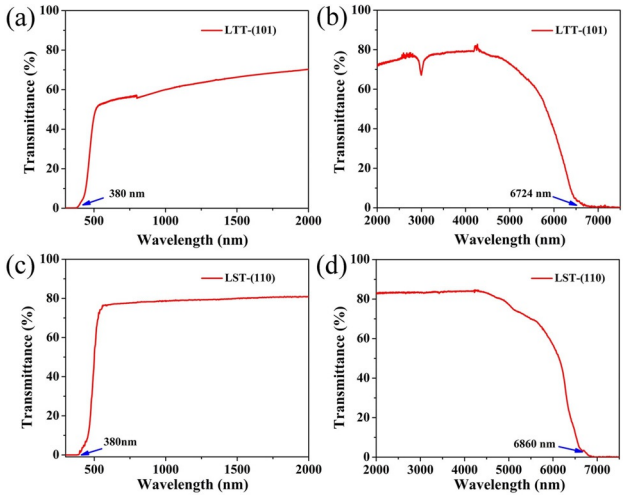

Figure 6. a,b) UV/Vis/IR transmission spectra of LTT crystals. c,d) UV/Vis/IR transmission spectra of LST crystals.

Figure S2, the band gap values of Li_2MTeO_6 (M = Ti, Sn) crystals are as large as 3.67 eV and 4.00 eV, implying that the two crystals may have considerable laser damage thresholds since a wide band gap generally leads to high laser damage thresholds.

The laser damage thresholds of LTT and LST crystals were measured at 1064 nm with a pump pulse width of 10 ns. LTT and LST crystals exhibit high laser-damage thresholds of > 550 MW cm^{-2} and > 672 MW cm^{-2}, respectively, which are higher than those of $AgGaS_2$ (30 MW cm^{-2}@1064 nm, 12 ns),[13] $LiNbO_3$ (60 MW cm^{-2}@1064 nm, 10 ns),[14] $ZnGeP_2$ (74 MW cm^{-2}@2050 nm, 10 ns),[13] and comparable with KTP (500 MW cm^{-2}@1064 nm, 10 ns),[15] $La_3SnGa_5O_{14}$ (846 MW cm^{-2}@1064 nm, 10 ns),[16] etc. With the optimization of the crystal quality, the LDTs of the two crystals would obviously be improved. The results indicate that the Li_2MTeO_6 (M = Ti, Sn) crystals have potential applications

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in the high-power mid-IR NLO region with excellent NLO properties and high LDT performance.

To further elucidate the structure and electronic origin of the optical properties of polar LTT and LST crystals, electronic structure calculations were done based on density functional theory (DFT) using the total-energy code Cambridge Sequential Total Energy Package (CASTEP). As illustrated in Figure 7, the band structures of LTT and LST crystals were calculated and analyzed.

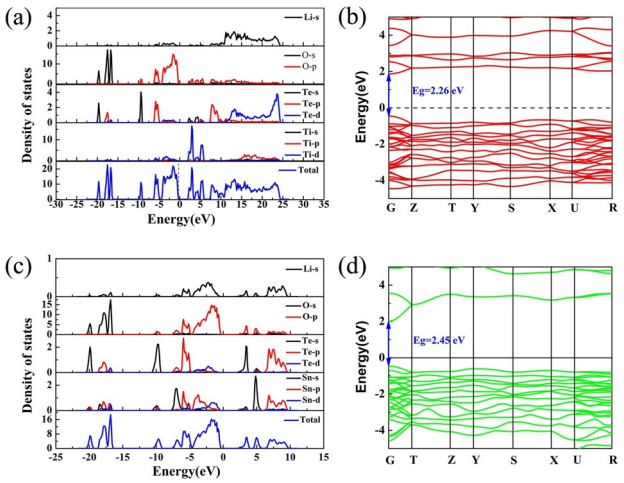

Figure 7. a) Band structure and b) TDOS and PDOS curves of LTT. c) Band structure and d) TDOS and PDOS curves of LST.

Both the highest valence band (VB) and the lowest conduction band (CB) of LTT and LST are located at the G point, which indicates the LTT and LST crystals are direct band gap materials. The band gap values were calculated to be 2.26 eV and 2.45 eV for LTT and LST, respectively. The generalized gradient approximation (GGA) generally underestimates the band gap,[17] therefore the theoretically determined band gap is usually smaller than the experimental one. On the basis of the total and partial densities of states (TDOS, PDOS, respectively) of the constituent atoms for LTT and LST crystals, the detailed electronic structure would be determined exactly, as shown in Figure 7 a,c.

Several electronic features of LTT can be deduced: (1) the region lower than −15 eV is composed of the Te 5s5p and O 2p orbitals; (2) the energy states between −10 eV and −9 eV are mainly dominated by Te 5s and O 2p orbitals; (3) the bottom of the VB is located in the energy range from 0 to −7 eV and the top of the CB near the Fermi level mostly consists of Ti 5p5d, Te 5p5d and O 2p orbitals with a small contribution from Te 5s orbitals. Some characteristics of LST can be deduced: (1) below −15 eV, the energy level is mainly dominated by the O 2s and Te 5s states, with very slight hybridization of Te 5p5d and Sn 4d5p orbitals; (2) the energy states between −10 eV and −9 eV consist of O 2p, Te 5s states with a small portion of Sn 5s5p states; (3) the orbitals that occupy the top of the valence band (−7.5–0 eV) and the bottom of the conduction band near the Fermi level (2.5–7 eV) are mainly composed of the O 2p, Te 5s5p, and Sn 5s5p orbitals, which plays a dominating role in the linear and nonlinear optical properties of the crystals. The contribution of the alkali-metal cations (Li^+) can be neglected in these states. According to these analyses, it is the TiO_6 octahedron and TeO_6 octahedron for LTT and the SnO_6 octahedron and TeO_6 octahedron for LST that make a considerable contribution to the optical properties of Li_2MTeO_6 (M = Ti, Sn) crystals.

Conclusion

Novel oxide crystals Li_2MTeO_6 (M = Ti, Sn) have been selected and grown as mid-IR nonlinear optical crystals. The LTT and LST crystals exhibit wide transmission ranges of 0.38–6.72 and 0.38–6.86 μm, respectively. The LTT crystal shows an intensive SHG response of 26 × KDP. The dipole moment calculation of NLO-active groups and first-principles analysis based on the crystal structures well explain the SHG response of LTT and LST. The results indicate that the LTT crystal would be an excellent nonlinear optical crystal with wide tranmission range and strong SHG response.

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Conflict of Interest

The authors declare no conflict of interest.

Keywords: laser damage threshold · Li_2SnTeO_6 · Li_2TiTeO_6 · mid-infrared · nonlinear optics

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